

A. Leave-One-Disaster-Out Cross-Validation Partition Matrix ($K = 6$)

	Train Pool	In-Domain Training Pool ($K - 1 = 5$ events)		Held-Out Test	Held-Out Test Set (Out-of-Domain Evaluation)	
	Ulysses (Nov 2020)	Rolly (Oct 2020)	Odette (Dec 2021)	Karding (Sep 2022)	Egay (Jul 2023)	Carina (Jul 2024)
Fold 1	Held-Out Test	Train Pool	Train Pool	Train Pool	Train Pool	Train Pool
Fold 2	Train Pool	Held-Out Test	Train Pool	Train Pool	Train Pool	Train Pool
Fold 3	Train Pool	Train Pool	Held-Out Test	Train Pool	Train Pool	Train Pool
Fold 4	Train Pool	Train Pool	Train Pool	Held-Out Test	Train Pool	Train Pool
Fold 5	Train Pool	Train Pool	Train Pool	Train Pool	Held-Out Test	Train Pool
Fold 6	Train Pool	Train Pool	Train Pool	Train Pool	Train Pool	Held-Out Test

Execution Protocol for Exemplar Fold k

B. Within-Fold Partitioning, Multi-Seed Optimization, and Checkpoint Selection

Training Pool ($K - 1 = 5$ Disaster Events)

- Combined corpus from five in-domain typhoons (~5,000 to 8,000 posts)
- Stratified partitioning executed once and held constant across models

85% Parameter Fitting

- Model weight updates
- Shared encoder optimization
- Task-specific loss gradients

15% Stratified Dev Split

- Stratified by event and tasks
- Checkpoint model selection
- Early stopping trigger (patience 5)

Multi-Seed Benchmarking Engine ($S = 5$ Random Seeds per Fold)

- Five random initialization seeds: $s \in \{42, 123, 456, 789, 1011\}$
- Six model configurations (identical compute budgets):
Single-Task: STL NER, STL Intent, STL Urgency
Multi-Task: MTL Uniform, MTL Tuned Static, MTL PCGrad
- Total experimental runs: 6 folds $\times$ 6 configurations $\times$ 5 seeds = 180 runs
- Checkpoint selection: State maximizing Macro-F1 on 15% dev split is frozen

Held-Out Disaster Event

- Strictly Unseen Out-of-Domain Set*
- 1 complete typhoon (~1,000 to 2,000 posts)
 - Zero gradient backpropagation
 - Zero hyperparameter tuning access
 - Evaluates true disaster transfer

Unseen Posts

Out-of-Domain Evaluation

- Evaluated per seed with frozen model
- Batch-size-one triage inference
- Seed metrics: $\{M_{k,s}\}_{s=1}^5$ across three tasks
- Evaluates all three triage heads

Best Model

Fold-Level Aggregation and Statistical Inference

C. Fold-Level Score Aggregation and Non-Parametric Permutation Hypothesis Testing

Fold-Level Seed Averaging

$$\bar{M}_k = \frac{1}{S} \sum_{s=1}^S M_{k,s} \quad (S = 5)$$

- Isolates optimization noise
- Prevents pseudo-replication
- Preserves fold as inferential unit

Paired Fold Observations ($N = K = 6$)

- Fold score vector per model:
 $\mathbf{m} = [M_1, \dots, M_6]$
- Paired difference across models:
 $d_k = M_{k, \text{Model A}} - \bar{M}_{k, \text{Model B}}$

Exact Paired Permutation Test

- Permutation count: $2^K = 2^6 = 64$
- Test statistic: $\bar{d} = \frac{1}{6} \sum_{k=1}^6 d_k$
- Minimum achievable one-sided p-value:
 $p_{\min} = 2^{-6} = .015625 < \alpha = .05$
- Free of distribution assumptions